Inside the Atalasoft dotImage ImageSource Class

Stephen Hawley, Chief Architect
Atalasoft, Inc.
December, 2005

THIS LOOKS COOL! CHECK IT OUT.

APPROVED

Abstract

Atalasoft dotImage includes many tools for manipulating or analyzing images. Included in the suite is a class called ImageSource which is the basis for being able to work effectively with an arbitrary number of images without worrying about the details of where they come from and how they're managed. This document will describe the ImageSource class and its descendents in detail.

REJECTED

In batch image processing or in image manipulation applications, it is often necessary to work with large quantities of images. While it is possible to write code to manage these for specific cases, it is often more convenient to be able to solve this problem in the more general case and then use that as a basis for more specific cases.

In dotImage, the abstract class ImageSource does just this. The way to think of ImageSource is to think of it as half of a source/sink pair. An ImageSource is a place from which images come. The sink is an application or other image consumer. Images are managed through an acquire/release model. The ImageSource object performs the following services:

- Allows images to be acquired in order
- Allows images to be released in any order
- Tracks memory used by images that were made available
- Automatically frees released images using either lazy or aggressive mechanisms
- Allows limited reacquisition of released images
- Allows for a reloading mechanism to allow images to be cached

In this model, an image can be thought of as a resource. Rather than simply being read and used, an image is acquired from the ImageSource and then released when finished. Any number of consumers can acquire any given image and it will only be released when each Acquire has been balanced with a Release.

In this way, an ImageSource can be used as follows:

```
public void ProcessImages(ImageSource source)
{
   while (source.HasMoreImages()) {
      AtalaImage image = source.AcquireNext();
      ProcessImage(image);
      source.Release(image);
   }
}
```

United States, that Matthes (1931) has adopted that name for the latest epoch in the Yosemite region. For the present, however, it seems best to retain a local name; and none of the names available seems more suitable than Tioga, for a glacier of this age occupied Tioga Pass and its lobes descended both southwest and northeast therefrom, leaving characteristic moraines and lakes. East slope localities where the moraines of this epoch are well displayed are Convict Lake, June Lake, Grant Lake, Leevining Canyon, Twin Lakes (Bridgeport Basin), Fallen Leaf Lake, and Donner Lake west of Truckee.

The glacial features that were made by the ice tongues of the Tioga epoch are even now almost as fresh and unaltered as at the time of their formation. The cirques are still as bare and ragged as if recently abandoned by the ice. Talus cones are few and small where the rocks are not closely jointed, and have not grown to large size even where the closer spacing of joints is favorable to frost action. The original polished and striated surface is still rather generally intact, even on such easily weathered rocks as coarse granite. Acres of polished and grooved rock are a familiar sight near Tenaya Lake (Yosemite National Park) and many other places in the high Sierra. If it were not for the destructive effects of forest fires, such surfaces would be even more completely preserved.

The lateral moraines generally stand out as bold embankments, marred only by a few landslides and by sharp ravines where tributary brooks descend the canyon sides. The terminal moraines are still complete, except for V-shaped notches through which the main streams tumble down to the plains beyond. As the distinctive glacial topography of the moraines is almost entirely preserved, it is generally an easy task to map them continuously.

Most of the valleys that were inhabited by glaciers of the fourth epoch now contain clear lakes 50 to 300 feet deep, some of which are a mile or more in length. The bouldery slopes commonly descend steeply into the water. Deltas at the upper ends of these lakes have grown forward only a small fraction of the original length of the lake, and the outlet streams have incised their morainic dams only 10–75 feet. Fallen Leaf Lake, near Lake Tahoe, held in by a compound loop of terminal moraines, is the largest lake of this type on the eastern slope of the range. Independence, Convict, and Donner lakes are similar. Only a few of the smaller and shallower lakes of this age have already disappeared. The rock-bound mountain tarns in the upper parts of the glacial troughs, largely surrounded by bare ice-scored rock mounds, are in substantially the same condition now as when the glaciers left them.

ATALASOFT DEVELOPER TOOLS LICENSE AGREEMENT

IMPORTANT - READ CAREFULLY: This License Agreement (the "Agreement") is a legal contract between you (either an individual or single business entity)("LICENSEE") and Atalasoft, Inc., ("Atalasoft" or "LICENSOR") for the software product identified above, which includes computer software and, as applicable, associated media, printed materials, and "online" or electronic documentation (collectively the "Software").

BY CLICKING THE AGREE OPTION BELOW, OR BY INSTALLING, COPYING OR OTHERWISE USING THE SOFTWARE, YOU AGREE TO BE BOUND BY THE TERMS OF THIS AGREEMENT, INCLUDING THE WARRANTY DISCLAIMERS, LIMITATIONS OF LIABILITY AND TERMINATION PROVISIONS BELOW. IF YOU DO NOT AGREE TO THE TERMS OF THIS AGREEMENT, DO NOT INSTALL THE SOFTWARE, AND EXIT NOW.

You are entitled to receive a full refund of any payment made for the Software by so requesting and returning any media and materials in unopened, unused condition and discarding any electronic serial numbers or registration information within 30 days after date of payment.

1. **LICENSE GRANT AND LICENSE TERMS.** This Agreement permits the LICENSEE the non-exclusive right to use one copy of a properly registered developer component, including but not limited to ImgX and dotImage, for development purposes on any single computer, provided the Software is in use on only one computer at any time. The Software is "in-use" on a computer when it is loaded into temporary memory (i.e. RAM) or installed into the permanent memory (i.e. hard disk, CD-ROM, or other storage device) of that computer. **The Software accompanying this Agreement, regardless of the media on which it is distributed, is licensed to LICENSEE by Atalasoft. LICENSEE owns the medium on which the Software is recorded, but Atalasoft or its licensors retain title to the Software and related documentation. The license granted under this Agreement is subject to the terms and conditions contained herein.**

   i. <u>One License Per Developer</u>. "Developer" is defined as an individual engaged in the development of a software application (the "Application"). Use of the Software requires one license per developer. Licenses are non-transferable between developers. For

RICHARD FOREMAN

SEQUELS OF THE WEEK

'Secret Games 2: The Escort' 'Chained Heat 2' 'Amityville: A New Generation' 'Lady Dragon 2' 'Relentless 3'

What 4?

UNATTRIBUTED QUOTE OF THE WEEK

"An over-the-top action thriller with hyper-kinetic edges"

From a trade magazine ad for *Billy Budd* (York, $89.95), about a murderous motorcyclist marauding through the Mojave.

ACTING STRETCH OF THE WEEK

Gary Burghoff in 'Small Kill'

(VCI, $59.95), in which *M*A*S*H*'s meek Radar O'Reilly plays a genius, homicidal, cross-dressing, psychotic drug lord wannabe.

ILLUSTRATION BY ERIC PALMER

women who advertise phone sex on TV are the ones who actually pick up calls. **D+** —*Doug Brod*

VINTAGE

DAVID HOLZMAN'S DIARY L.M. Kit Carson *(1967, Fox Lorber, unrated, $79.95)* Years before he went Hollywood with *The Big Easy* and *Great Balls of Fire!*, director Jim McBride scored with this droll put-on, a fake underground movie that shoots snarky darts through film-student pretensions. David Holzman (Carson) is a sincere young twerp who, seeking Big Truth, films his life in what was the then-fashionable grainy B&W vérité. What he finds is a swift corrective to all those who think movies are more important than reality. Raw, ragged, and right, *Diary* is both an unerring late-'60s time capsule and as modern as teen angst. **A-** —*TB*

DOCUMENTARY

THE LOST FLEET OF GUADALCANAL *(1993, Columbia TriStar, $19.95)* When Robert Ballard, who discovered the *Titanic* and the *Bismarck*, set off in 1992 to find 50 "ghost ships" downed in the World War II battle of Guadalcanal, he met with only modest success. Fortunately, he'd brought reinforcements: American and Japanese survivors of the harrowing six-month fight. Their moving reminiscences 50 years later, accompanied by vivid archival footage, far eclipse Ballard's underwater video of sunken metal. Combining elements of the Jacques Cousteau and war-documentary genres, National Geographic's *Lost Fleet* ultimately transcends both. **A** —*Susan Chumsky*

MUSIC

PRIDE AND JOY: THE STORY OF ALLIGATOR RECORDS Koko Taylor, Lonnie Brooks *(1993, BMG, $29.95)* For more than two decades, Alligator Records has been introducing young, mostly white music fans to contemporary hard blues. The label's success story—captured in this 87-minute documentary—involves equal parts nostalgia, clever marketing, and a genuine labor of love. Director Robert Mugge (*Deep Blues*) places Alligator in the context of the Chicago blues scene, interviews key players, and presents some enthralling live performances. But the tape still suffers from the perfunctory: In band interviews, Alligator artists reveal a good-natured sincerity without ever digging deeper than Taylor's remark, "We ate it. We slept it. We was born with the blues." **C+** —*Taehee Kim*

TOP VIDEOS

'CRYING' TIME

WITH ARTSIER video renters having run through *Howards End* (which drops off the top 10 this week), *The Crying Game* finally joins the chart at No. 7 in its third week in release. Apparently, laserdisc buyers were awaiting the Oscar nominee far more eagerly than tape renters: The laser version debuts at a surprising No. 2.

Meanwhile, the upward mobility of Columbia TriStar/Voyager's laserdisc of *Bram Stoker's Dracula* may soon be halted. Two just-released inexpensive, wide-screen versions should cut into the sales of the current pricey ($124.99), feature-packed edition.

TOP 10 TAPE RENTALS

	LAST CHART		WEEKS ON CHART
1	1	**UNFORGIVEN** Clint Eastwood, *Warner*	3
2	5	**THE BODYGUARD** Kevin Costner, *Warner*	2
3	2	**A FEW GOOD MEN** Tom Cruise, *Columbia TriStar*	4
4	3	**BRAM STOKER'S DRACULA** Gary Oldman, *Columbia TriStar*	5
5	4	**LEAP OF FAITH** Steve Martin, *Paramount*	4
6	9	**TEENAGE MUTANT NINJA TURTLES III** Elias Koteas, *New Line*	2
7	12	**THE CRYING GAME** Stephen Rea, *LIVE*	3
8	6	**FOREVER YOUNG** Mel Gibson, *Warner*	7
9	11	**LORENZO'S OIL** Susan Sarandon, *MCA/Universal*	2
10	7	**JENNIFER 8** Andy Garcia, *Paramount*	5

SALES

1	1	**PINOCCHIO** Animated, *Disney, $24.99*	18
2	—	**TEENAGE MUTANT NINJA TURTLES III** Elias Koteas, *New Line, $22.99*	1
3	2	**SING-ALONG SONGS: FRIEND LIKE ME** Animated, *Disney, $12.99*	13
4	4	**RICHARD SIMMONS: SWEATIN' TO THE OLDIES 2** Fitness, *GoodTimes, $19.98*	28
5	3	**BEAUTY AND THE BEAST** Animated, *Disney, $24.99*	39
6	7	**BARNEY GOES TO SCHOOL** Barney, *Lyons, $14.95*	22
7	8	**DANCES WITH WOLVES** Kevin Costner, *Orion, $14.98*	18
8	10	**THE LITTLE MERMAID: STORMY, THE WILD SEAHORSE** Animated, *Disney, $12.99*	22
9	9	**BARNEY'S BEST MANNERS** Barney, *Lyons, $14.95*	16
10	6	**101 DALMATIANS** Animated, *Disney, $24.99*	68

LASERDISC SALES

1	—	**UNFORGIVEN** Clint Eastwood, *Warner, $39.98*	1
2	—	**THE CRYING GAME** Stephen Rea, *LIVE, $34.95*	1
3	1	**BLADE RUNNER** Harrison Ford, *Warner, $49.98*	17
4	7	**BRAM STOKER'S DRACULA** Gary Oldman, *Columbia TriStar/Voyager, $124.99*	3
5	3	**BODY OF EVIDENCE** Madonna, *MGM/UA, $34.98*	5
6	2	**HOWARDS END** Emma Thompson, *Columbia TriStar, $39.95*	7
7	4	**UNDER SIEGE** Steven Seagal, *Warner, $29.98*	13
8	8	**THE LAST OF THE MOHICANS** Daniel Day-Lewis, *FoxVideo/Image, $39.98*	15
9	—	**MATINEE** John Goodman, *MCA/Universal, $34.98*	1
10	18	**JENNIFER 8** Andy Garcia, *Paramount, $39.95*	3

TAPE DATA: *VIDEO BUSINESS* FOR THE WEEK ENDING JULY 25, 1993
LASERDISC DATA: *BILLBOARD* FOR THE WEEK ENDING JULY 31, 1993

Value Added Tax Return

HM Customs and Excise

For the period
01 03 91 to 30 04 93

For Official Use

099 777 3333 05 100 04 93 C03615
THE LLANFAIRPWLLGWYNRYNGOGERYCHLLA-
NDYSILIOGOGOGOCH SOCIETY FOR THE
PREVENTION OF ABBREVIATIONS
SOMEWHEREIN
WALES
GREAT BRITAIN
WWWW1234 008597/10

Registration Number: 777 3333 05

Period: 04 93

You could be liable to a financial penalty if your completed return and all the VAT payable are not received by the due date.

Due date: 31 05 93

For Official Use: ENTERED ON COMPUTER

Your VAT Office telephone number is 0702 367720

ATTENTION

If you are using Retail Scheme B1, D or J, please remember to carry out your annual adjustment at the appropriate time.

Before you fill in this form please read the notes on the back and the VAT leaflet *"Filling in your VAT return"*. Fill in all boxes clearly in ink, and write 'none' where necessary. Don't put a dash or leave any box blank. If there are no pence write **"00"** in the pence column. **Do not** enter more than one amount in any box.

For official use

		£	p
VAT due in this period on sales and other outputs	1		
VAT due in this period on **acquisitions** from other **EC Member States**	2		
Total VAT due **(the sum of boxes 1 and 2)**	3		
VAT reclaimed in this period on **purchases** and other inputs (including acquisitions from the EC)	4		
Net VAT to be paid to Customs or reclaimed by you (Difference between boxes 3 and 4)	5		
Total value of **sales** and all other outputs excluding any VAT. **Include your box 8 figure**	6		00
Total value of **purchases** and all other inputs excluding any VAT. **Include your box 9 figure**	7		00
Total value of all **supplies** of goods and related services, excluding any VAT, to other **EC Member States**	8		00
Total value of all **acquisitions** of goods and related services, excluding any VAT, from other **EC Member States**	9	200 00	00

Retail schemes. If you have used any of the schemes in the period covered by this return, enter the relevant letter(s) in this box.

If you are enclosing a payment please tick this box. ☐

DECLARATION: You, or someone on your behalf, must sign below.

I, ...Jon Doe.... (Full name of signatory in BLOCK LETTERS) ....declare that the information given above is true and complete.

Signature ....Jon Doe.... Date .....19.....

A false declaration can result in prosecution.

B

CD 2859/N3(04/92) F3790 (JANUARY 1993)

VAT 100

Your Company Name

Your Company Slogan

TIME SHEET

Address
City, State ZIP
123.456.7890 Fax 123.456.7891

Employee Name: John Doe

Title: Programmer

Employee Number: 123456789

Status:

Department: Engineering

Supervisor: William Clinton

Date	Start Time	End Time	Regular Hrs.	Overtime Hrs.	Total Hrs.
1/3/06	8:00 AM	5:00 PM	8	0	8
1/4/06	7:00 AM	6:00 PM	10	2	10
1/11/06	8:00 AM	5:00 PM	8	0	8
		WEEKLY TOTALS:	27	2	29

Employee Signature: John Doe

Date: 1/11/06

Supervisor Signature: William B.

Date: